

INF2003: Database Systems

Lab 03: Complex SQL Queries

Preparation Step

Download and run the SQL script (sample_data.sql) for the creation of the relational schema and insertion of data from LMS.

[illegible][illegible]

```
CREATE DATABASE Johndb;  
USE Johndb;  
SOURCE /Users/aungsh/Documents/school/Year 2 Tri 1/INF2003-Database Systems/Lab/lab-3/sample_data.sql;  
SHOW TABLES;
```

Exercise: Queries in SQL.

(Note: Remember to save your work and screen capture your SQL queries and output for your lab submission.)

Q1. Find the emails of students.

[illegible]

```
SELECT email FROM student;
```

Q2. Find the different emails of students. (Do we need to use DISTINCT?)

[illegible]

```
SELECT DISTINCT email FROM student;
```

Q3. Print the names of students in descending alphabetical order.

[illegible]

```
SELECT name FROM student ORDER BY name DESC;
```

SIT Restricted

Q4. Find out if there are students with the same name?

[illegible]

```
SELECT name, COUNT GROUP BY name HAVING COUNT(*) > 1;
```

Q5. Find the different names of students. Is the result sorted? If not, please sort it.

[illegible]

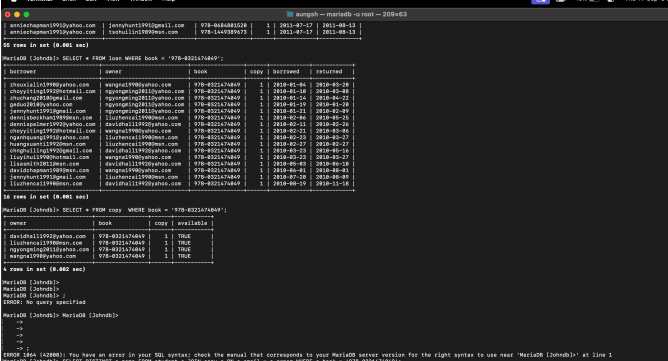
```
SELECT DISTINCT name FROM student ORDER BY name ASC;
```

Q6. Find the names of students who owned a copy of the book '978-0321474049'.

```

Terminal Shell Edit View Window Help







```

```
SELECT DISTINCT s.name FROM student s JOIN copy c ON s.email = c.owner WHERE c.book = '978-0321474049';
```

Q7. Find the names of the students who owned a copy of a book with more than 100 pages and the book title contains the word 'Photoshop'.

```

Terminal Shell Edit View Window Help
aungah -- mariadb -u root -- 200x63

davidhall11992@yahoo.com | 978-0321474049 | 1 | TRUE |
liuhencai1980@gmail.com | 978-0321474049 | 1 | TRUE |
ngyongting81@yahoo.com | 978-0321474049 | 1 | TRUE |
wangna1990@yahoo.com | 978-0321474049 | 1 | TRUE |
4 rows in set (0.002 sec)

MariaDB [johndb]>
MariaDB [johndb]>
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near 'MariaDB [johndb]>' at line 1
MariaDB [johndb]> SELECT DISTINCT s.name FROM student s JOIN copy c ON s.email = c.owner WHERE c.book = '978-0321474049';
+-----+
| name |
+-----+
| DAVID HALL |
| LIU ZHENCAI |
| NG YONG TING |
| WANG NA |
+-----+
4 rows in set (0.003 sec)

MariaDB [johndb]> | WANG NA |
+-----+
| WANG NA |
+-----+
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near 'WANG NA |' at line 1
MariaDB [johndb]> SELECT * FROM book;
+-----+-----+-----+-----+-----+-----+-----+-----+
| title | format | pages | language | authors | publisher | year | ISBN10 | ISBN13 |
+-----+-----+-----+-----+-----+-----+-----+-----+
| The Digital Photography Book | paperback | 219 | English | Scott Kelly | Peachpit Press | 2006-01-01 | 0321474049 | 978-0321474049 |
| The Great Gatsby | hardcover | 216 | English | F. Scott Fitzgerald | Scribner | 1998-01-01 | 0684801520 | 978-0684801520 |
| Photoshop Elements 9: The Missing Manual | paperback | 640 | English | Barbara Brundage | Pogue Press | 2010-01-01 | 1449389678 | 978-1449389678 |
| Where Good Ideas Come From: The Natural History of Innovation | hardcover | 336 | English | Steven Johnson | Riverhead Hardcover | 2010-01-01 | 1594487712 | 978-1594487712 |
+-----+-----+-----+-----+-----+-----+-----+-----+
4 rows in set (0.001 sec)

MariaDB [johndb]> SELECT DISTINCT s.name FROM student JOIN copy c ON s.email = c.owner;MariaDB [johndb]> SELECT DISTINCT s.name FROM student JOIN copy c ON 'Photoshop';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near 'LIKE'Photoshop'' at line 1
MariaDB [johndb]> SELECT DISTINCT s.name FROM student JOIN copy c ON s.email = c.owner JOIN book b ON c.book = b.ISBN13 WHERE b.pages > 100 AND b.title LIKE 'Photoshop%';
ERROR 1064 (42000): unknown column 's.name' in 'SELECT'
MariaDB [johndb]> SELECT DISTINCT s.name FROM student s JOIN copy c ON s.email = c.owner JOIN book b ON c.book = b.ISBN13 WHERE b.pages > 100 AND b.title LIKE 'Photoshop%';
+-----+
| name |
+-----+
| GUN HUI YING |
| JENNY FANG |
| TSO HUI LIN |
| WANG NA |
+-----+
4 rows in set (0.002 sec)

MariaDB [johndb]>

```

SELECT DISTINCT s.name FROM student s JOIN copy c ON s.email = c.owner JOIN book b ON c.book = b.ISBN13 WHERE b.pages > 100 AND b.title LIKE '%Photoshop%';

Q8. Find the number of A4 paper needed to photocopy the two books with ISBN-13 '978-0321474049' and '978-0684801520' (duplex printing, or 2 pages of a book can be copied on one A4 paper).

```

Terminal Shell Edit View Window Help
aungah -- mariadb -u root -- 200x63

MariaDB [johndb]>
MariaDB [johndb]>
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near 'MariaDB [johndb]>' at line 1
MariaDB [johndb]> SELECT DISTINCT s.name FROM student s JOIN copy c ON s.email = c.owner WHERE c.book = '978-0321474049';
+-----+
| name |
+-----+
| DAVID HALL |
| LIU ZHENCAI |
| NG YONG TING |
| WANG NA |
+-----+
4 rows in set (0.002 sec)

MariaDB [johndb]> | WANG NA |
+-----+
| WANG NA |
+-----+
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near 'WANG NA |' at line 1
MariaDB [johndb]> SELECT * FROM book;
+-----+-----+-----+-----+-----+-----+-----+-----+
| title | format | pages | language | authors | publisher | year | ISBN10 | ISBN13 |
+-----+-----+-----+-----+-----+-----+-----+-----+
| The Digital Photography Book | paperback | 219 | English | Scott Kelly | Peachpit Press | 2006-01-01 | 0321474049 | 978-0321474049 |
| The Great Gatsby | hardcover | 216 | English | F. Scott Fitzgerald | Scribner | 1998-01-01 | 0684801520 | 978-0684801520 |
| Photoshop Elements 9: The Missing Manual | paperback | 640 | English | Barbara Brundage | Pogue Press | 2010-01-01 | 1449389678 | 978-1449389678 |
| Where Good Ideas Come From: The Natural History of Innovation | hardcover | 336 | English | Steven Johnson | Riverhead Hardcover | 2010-01-01 | 1594487712 | 978-1594487712 |
+-----+-----+-----+-----+-----+-----+-----+-----+
4 rows in set (0.001 sec)

MariaDB [johndb]> SELECT DISTINCT s.name FROM student JOIN copy c ON s.email = c.owner;MariaDB [johndb]> SELECT DISTINCT s.name FROM student JOIN copy c ON 'Photoshop';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near 'LIKE'Photoshop'' at line 1
MariaDB [johndb]> SELECT DISTINCT s.name FROM student JOIN copy c ON s.email = c.owner JOIN book b ON c.book = b.ISBN13 WHERE b.pages > 100 AND b.title LIKE 'Photoshop';
ERROR 1064 (42000): unknown column 's.name' in 'SELECT'
MariaDB [johndb]> SELECT DISTINCT s.name FROM student s JOIN copy c ON s.email = c.owner JOIN book b ON c.book = b.ISBN13 WHERE b.pages > 100 AND b.title LIKE 'Photoshop';
+-----+
| name |
+-----+
| GUN HUI YING |
| JENNY FANG |
| TSO HUI LIN |
| WANG NA |
+-----+
4 rows in set (0.002 sec)

MariaDB [johndb]> SELECT CEIL(SUM(pages)/2) AS a4_papers_needed FROM book WHERE ISBN13 IN ('978-0321474049', '978-0684801520');
+-----+
| a4_papers_needed |
+-----+
| 218 |
+-----+
1 row in set (0.002 sec)

MariaDB [johndb]>

```

SELECT CEIL(SUM(pages)/2) AS a4_papers_needed FROM book WHERE ISBN13 IN ('978-0321474049', '978-0684801520');

Q9. Find the different names of the students who owned a copy of a book other than '978-0321474049'. (Hint: a student may own multiple books)

```

Terminal Shell Edit View Window Help
mangsh-mariadb-root-200463

11:20:42Z
MariaDB [(none)]>
MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| information |
| mysql |
| performance_schema |
| test |
+-----+

MariaDB [(none)]> use information;
MariaDB [(none)]> show tables;
+-----+
| Tables_in_information |
+-----+
| books |
| student |
| student_loan |
+-----+

MariaDB [(none)]> describe books;
+-----+
| Field | Type | Null | Key | Extra |
+-----+
| title | varchar(255) | NO | PRIMARY |  |
| isbn | varchar(15) | NO | PRIMARY |  |
| author | varchar(255) | NO |  |  |
| publisher | varchar(255) | NO |  |  |
| year | int(4) | NO |  |  |
+-----+

MariaDB [(none)]> describe student;
+-----+
| Field | Type | Null | Key | Extra |
+-----+
| sname | varchar(255) | NO | PRIMARY |  |
| email | varchar(255) | NO |  |  |
| owner | varchar(255) | NO |  |  |
| book | varchar(255) | NO |  |  |
+-----+

MariaDB [(none)]> describe student_loan;
+-----+
| Field | Type | Null | Key | Extra |
+-----+
| sl_id | int(4) | NO | PRIMARY |  |
| sname | varchar(255) | NO |  |  |
| email | varchar(255) | NO |  |  |
| owner | varchar(255) | NO |  |  |
| book | varchar(255) | NO |  |  |
| borrow | int(4) | NO |  |  |
+-----+

MariaDB [(none)]> select * from books;
+-----+
| title | isbn | author | publisher | year |
+-----+
| The Digital Photography Book | 978-0321474049 | Scott Kelly | No Starch Press | 2004 |
| The Small Screen | 978-0321474049 | J. Scott Fitzgerald | No Starch Press | 2004 |
| The Small Screen | 978-0321474049 | J. Scott Fitzgerald | No Starch Press | 2004 |
| The Small Screen | 978-0321474049 | J. Scott Fitzgerald | No Starch Press | 2004 |
+-----+

MariaDB [(none)]> select * from student;
+-----+
| sname | email | owner | book |
+-----+
| David | david@no-starch.com | David | 978-0321474049 |
| David | david@no-starch.com | David | 978-0321474049 |
| David | david@no-starch.com | David | 978-0321474049 |
| David | david@no-starch.com | David | 978-0321474049 |
+-----+

MariaDB [(none)]> select * from student_loan;
+-----+
| sl_id | sname | email | owner | book | borrow |
+-----+
| 1 | David | david@no-starch.com | David | 978-0321474049 | 1 |
| 2 | David | david@no-starch.com | David | 978-0321474049 | 1 |
| 3 | David | david@no-starch.com | David | 978-0321474049 | 1 |
| 4 | David | david@no-starch.com | David | 978-0321474049 | 1 |
+-----+

MariaDB [(none)]>

```

SELECT DISTINCT s.name FROM student s JOIN copy c ON s.email = c.owner WHERE c.book <> '978-0321474049'

Q10. Find the names of students who borrowed a copy of the book '978-0321474049'.

<YOUR INPUT HERE>

```

Terminal Shell Edit View Window Help
mangsh-mariadb-root-200463

11:20:42Z
MariaDB [(none)]>
MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| information |
| mysql |
| performance_schema |
| test |
+-----+

MariaDB [(none)]> use information;
MariaDB [(none)]> show tables;
+-----+
| Tables_in_information |
+-----+
| books |
| student |
| student_loan |
+-----+

MariaDB [(none)]> describe books;
+-----+
| Field | Type | Null | Key | Extra |
+-----+
| title | varchar(255) | NO | PRIMARY |  |
| isbn | varchar(15) | NO | PRIMARY |  |
| author | varchar(255) | NO |  |  |
| publisher | varchar(255) | NO |  |  |
| year | int(4) | NO |  |  |
+-----+

MariaDB [(none)]> describe student;
+-----+
| Field | Type | Null | Key | Extra |
+-----+
| sname | varchar(255) | NO | PRIMARY |  |
| email | varchar(255) | NO |  |  |
| owner | varchar(255) | NO |  |  |
| book | varchar(255) | NO |  |  |
+-----+

MariaDB [(none)]> describe student_loan;
+-----+
| Field | Type | Null | Key | Extra |
+-----+
| sl_id | int(4) | NO | PRIMARY |  |
| sname | varchar(255) | NO |  |  |
| email | varchar(255) | NO |  |  |
| owner | varchar(255) | NO |  |  |
| book | varchar(255) | NO |  |  |
| borrow | int(4) | NO |  |  |
+-----+

MariaDB [(none)]> select * from books;
+-----+
| title | isbn | author | publisher | year |
+-----+
| The Digital Photography Book | 978-0321474049 | Scott Kelly | No Starch Press | 2004 |
| The Small Screen | 978-0321474049 | J. Scott Fitzgerald | No Starch Press | 2004 |
| The Small Screen | 978-0321474049 | J. Scott Fitzgerald | No Starch Press | 2004 |
| The Small Screen | 978-0321474049 | J. Scott Fitzgerald | No Starch Press | 2004 |
+-----+

MariaDB [(none)]> select * from student;
+-----+
| sname | email | owner | book |
+-----+
| David | david@no-starch.com | David | 978-0321474049 |
| David | david@no-starch.com | David | 978-0321474049 |
| David | david@no-starch.com | David | 978-0321474049 |
| David | david@no-starch.com | David | 978-0321474049 |
+-----+

MariaDB [(none)]> select * from student_loan;
+-----+
| sl_id | sname | email | owner | book | borrow |
+-----+
| 1 | David | david@no-starch.com | David | 978-0321474049 | 1 |
| 2 | David | david@no-starch.com | David | 978-0321474049 | 1 |
| 3 | David | david@no-starch.com | David | 978-0321474049 | 1 |
| 4 | David | david@no-starch.com | David | 978-0321474049 | 1 |
+-----+

MariaDB [(none)]>

```

SELECT DISTINCT s.name FROM student s JOIN loan l ON s.email = l.borrower WHERE l.book = '978-0321474049';

Q11. Find the names of students who owned or borrowed a copy of the book '978-0321474049' using set operator UNION.

```

Terminal Shell Edit View Window Help
mangsh-mariadb-root-200463

11:20:42Z
MariaDB [(none)]>
MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| information |
| mysql |
| performance_schema |
| test |
+-----+

MariaDB [(none)]> use information;
MariaDB [(none)]> show tables;
+-----+
| Tables_in_information |
+-----+
| books |
| student |
| student_loan |
+-----+

MariaDB [(none)]> describe books;
+-----+
| Field | Type | Null | Key | Extra |
+-----+
| title | varchar(255) | NO | PRIMARY |  |
| isbn | varchar(15) | NO | PRIMARY |  |
| author | varchar(255) | NO |  |  |
| publisher | varchar(255) | NO |  |  |
| year | int(4) | NO |  |  |
+-----+

MariaDB [(none)]> describe student;
+-----+
| Field | Type | Null | Key | Extra |
+-----+
| sname | varchar(255) | NO | PRIMARY |  |
| email | varchar(255) | NO |  |  |
| owner | varchar(255) | NO |  |  |
| book | varchar(255) | NO |  |  |
+-----+

MariaDB [(none)]> describe student_loan;
+-----+
| Field | Type | Null | Key | Extra |
+-----+
| sl_id | int(4) | NO | PRIMARY |  |
| sname | varchar(255) | NO |  |  |
| email | varchar(255) | NO |  |  |
| owner | varchar(255) | NO |  |  |
| book | varchar(255) | NO |  |  |
| borrow | int(4) | NO |  |  |
+-----+

MariaDB [(none)]> select * from books;
+-----+
| title | isbn | author | publisher | year |
+-----+
| The Digital Photography Book | 978-0321474049 | Scott Kelly | No Starch Press | 2004 |
| The Small Screen | 978-0321474049 | J. Scott Fitzgerald | No Starch Press | 2004 |
| The Small Screen | 978-0321474049 | J. Scott Fitzgerald | No Starch Press | 2004 |
| The Small Screen | 978-0321474049 | J. Scott Fitzgerald | No Starch Press | 2004 |
+-----+

MariaDB [(none)]> select * from student;
+-----+
| sname | email | owner | book |
+-----+
| David | david@no-starch.com | David | 978-0321474049 |
| David | david@no-starch.com | David | 978-0321474049 |
| David | david@no-starch.com | David | 978-0321474049 |
| David | david@no-starch.com | David | 978-0321474049 |
+-----+

MariaDB [(none)]> select * from student_loan;
+-----+
| sl_id | sname | email | owner | book | borrow |
+-----+
| 1 | David | david@no-starch.com | David | 978-0321474049 | 1 |
| 2 | David | david@no-starch.com | David | 978-0321474049 | 1 |
| 3 | David | david@no-starch.com | David | 978-0321474049 | 1 |
| 4 | David | david@no-starch.com | David | 978-0321474049 | 1 |
+-----+

MariaDB [(none)]>

```

SELECT s.name FROM student s JOIN copy c ON s.email = c.owner WHERE c.book = '978-0321474049' UNION SELECT s.name FROM student s JOIN loan l ON s.email = l.borrower WHERE l.book = '978-0321474049';

Q12. Find the names of students who owned or borrowed a copy of the book '978-0321474049' using logical operator OR.

```

Terminal Shell Edit View Window Help
aungsh - mariadb -u root - 209x63

DENNIS PALMER |
GE DUO       |
HUANG XIANZI |
JENNY HUNT   |
LISA SMITH   |
LIU YIHUI    |
LIU ZHENCAI  |
NG ANH QUANG |
ZHOU XIALIN  |
ZHU CHANG    |
+-----+
34 rows in set (0.001 sec)

MariaDB [johndb]> SELECT s.name FROM student s JOIN copy c ON s.email = c.owner WHERE c.book = '978-0321474049' UNION SELECT s.name FROM student s JOIN loan l ON s.email = l.borrower WHERE l.book = '978-0321474049';
+-----+
| name |
+-----+
| DAVID HALL |
| LIU ZHENCAI |
| NG YONG MING |
| WANG NA |
| CHING HUI LING |
| CHOI YI TING |
| DAVID CHAPMAN |
| DENNIS BECKHAM |
| DENNIS PALMER |
| GE DUO |
| HUANG XIANZI |
| JENNY HUNT |
| LISA SMITH |
| LIU YIHUI |
| LIU ZHENCAI |
| NG ANH QUANG |
| NG YONG MING |
| WANG NA |
| ZHOU XIALIN |
| ZHU CHANG |
+-----+
37 rows in set (0.002 sec)

MariaDB [johndb]> SELECT DISTINCT name FROM student WHERE email IN (SELECT owner FROM copy WHERE book = '978-0321474049') OR email IN (SELECT borrower FROM loan WHERE BOOK = '978-0321474049');
+-----+
| name |
+-----+
| CHING HUI LING |
| CHOI YI TING |
| DAVID CHAPMAN |
| DAVID HALL |
| DENNIS BECKHAM |
| DENNIS PALMER |
| GE DUO |
| HUANG XIANZI |
| JENNY HUNT |
| LISA SMITH |
| LIU YIHUI |
| LIU ZHENCAI |
| NG ANH QUANG |
| NG YONG MING |
| WANG NA |
| ZHOU XIALIN |
| ZHU CHANG |
+-----+
37 rows in set (0.001 sec)

MariaDB [johndb]>

```

SELECT DISTINCT name FROM student WHERE email IN (SELECT owner FROM copy WHERE book = '978-0321474049') OR email IN (SELECT borrower FROM loan WHERE BOOK = '978-0321474049');

Q13. Delete all the data in table loan. (Hint: do not delete the table)

Try the Q12 query again, and comment on the results.

```

Terminal Shell Edit View Window Help
aungsh - mariadb -u root - 209x63

+-----+
| name |
+-----+
| DAVID HALL |
| LIU ZHENCAI |
| NG YONG MING |
| WANG NA |
| CHING HUI LING |
| CHOI YI TING |
| DAVID CHAPMAN |
| DENNIS BECKHAM |
| DENNIS PALMER |
| GE DUO |
| HUANG XIANZI |
| JENNY HUNT |
| LISA SMITH |
| LIU YIHUI |
| LIU ZHENCAI |
| NG ANH QUANG |
| NG YONG MING |
| WANG NA |
| ZHOU XIALIN |
| ZHU CHANG |
+-----+
37 rows in set (0.002 sec)

MariaDB [johndb]> SELECT DISTINCT name FROM student WHERE email IN (SELECT owner FROM copy WHERE book = '978-0321474049') OR email IN (SELECT borrower FROM loan WHERE BOOK = '978-0321474049');
+-----+
| name |
+-----+
| CHING HUI LING |
| CHOI YI TING |
| DAVID CHAPMAN |
| DAVID HALL |
| DENNIS BECKHAM |
| DENNIS PALMER |
| GE DUO |
| HUANG XIANZI |
| JENNY HUNT |
| LISA SMITH |
| LIU YIHUI |
| LIU ZHENCAI |
| NG ANH QUANG |
| NG YONG MING |
| WANG NA |
| ZHOU XIALIN |
| ZHU CHANG |
+-----+
37 rows in set (0.001 sec)

MariaDB [johndb]> DELETE FROM loan;
--
Query OK, 55 rows affected (0.001 sec)

MariaDB [johndb]> SELECT DISTINCT name FROM student WHERE email IN (SELECT owner FROM copy WHERE book = '978-0321474049') OR email IN (SELECT borrower FROM loan WHERE BOOK = '978-0321474049');
+-----+
| name |
+-----+
| DAVID HALL |
| LIU ZHENCAI |
| NG YONG MING |
| WANG NA |
+-----+
4 rows in set (0.002 sec)

MariaDB [johndb]>

```

DELETE FROM loan;

SELECT DISTINCT name FROM student WHERE email IN (SELECT owner FROM copy WHERE book = '978-0321474049') OR email IN (SELECT borrower FROM loan WHERE BOOK = '978-0321474049');

Only the students who owned the book are returned. Since the loan table is empty, no borrowers are found, but the owners are preserved.